



HITEC UNIVERSITY TAXILA CANTT
BS CYBER SECURITY
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AI LAB REPORT:4

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EMPLOYEE MANAGEMENT

- Create a Person class with attributes/variables “name” and “age”.
- Create a Job class with attributes/variables “designation” and “salary”.
- Create an Employee class that inherits the above both classes
- Implement a method to display employee details.

CODE:

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

# Job class
class Job:
    def __init__(self, designation, salary):
        self.designation = designation
        self.salary = salary

# Employee class inheriting from both Person and Job
class Employee(Person, Job):
    def __init__(self, name, age, designation, salary):
        Person.__init__(self, name, age)
        Job.__init__(self, designation, salary)

    def display_details(self):
        print("---- Employee Details ----")
        print("Name:", self.name)
        print("Age:", self.age)
        print("Designation:", self.designation)
        print("Salary:", self.salary)

# Example:
emp1 = Employee("HALIMA", 19, "Software Engineer", 75000)
emp1.display_details()
```

OUTPUT:

```
Name: HALIMA
Age: 19
Designation: Software Engineer
Salary: 75000
PS C:\Users\HP\Documents\New folder>
```

QUESTION NO:1

Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

CODE:

```
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * self.radius ** 2

    def perimeter(self):
        return 2 * math.pi * self.radius

# Example:
r = float(input("Enter the radius of the circle: "))
circle = Circle(r)
print("Area of circle:", circle.area())
print("Perimeter of circle:", circle.perimeter())
```

OUTPUT:

```
python3 debug.py 20251111 11:52:40 AM
Enter the radius of the circle: 6
Area of circle: 113.09733552923255
Perimeter of circle: 37.69911184307752
PS C:\Users\HP\Documents\New folder>
```

QUESTION NO:2

Write a Python class to implement $\text{pow}(x, n)$. Do not use built-in `math.pow()` function for this code. Also handle this code for all +ve as well as -ve numbers.

CODE:

```
class Power:
    def pow(self, x, n):
        result = 1
        if n > 0:
            for i in range(n):
                result *= x
        elif n < 0:
            for i in range(-n):
                result *= x
            result = 1 / result
        # if n == 0, result remains 1
        return result

# Example:
x = float(input("Enter the base number (x): "))
n = int(input("Enter the exponent (n): "))
p = Power()
print(f"{x} raised to the power {n} is:", p.pow(x, n))
```

OUTPUT:

```
Enter the base number (x): 5
Enter the exponent (n): 5
5.0 raised to the power 5 is: 3125.0
PS C:\Users\HP\Documents\New folder> |
```

QUESTION NO:3

```
class Item:
    def __init__(self, name, description, price):
        self.name = name
        self.description = description
        self.price = price

    def viewFullDescription(self):
        print("Item Name:", self.name)
```

```

        print("Description:", self.description)
        print("Price:", self.price)

# Derived class: MP3
class MP3(Item):
    def __init__(self, name, description, price, artist, duration):
        super().__init__(name, description, price)
        self.artist = artist
        self.duration = duration

    def play(self):
        print("Playing:", self.name, "by", self.artist)

# Derived class: DVD
class DVD(Item):
    def __init__(self, name, description, price, certificate, duration, actors):
        super().__init__(name, description, price)
        self.certificate = certificate
        self.duration = duration
        self.actors = actors

    def viewTrailer(self):
        print("Viewing trailer of:", self.name)
        print("Actors:", ", ".join(self.actors))

# Derived class: Book
class Book(Item):
    def __init__(self, name, description, price, author, pages, genre):
        super().__init__(name, description, price)
        self.author = author
        self.pages = pages
        self.genre = genre

    def previewContent(self):
        print("Previewing book:", self.name, "by", self.author)

# Pakistani Data Examples
mp3 = MP3("Tera Woh Pyar", "Coke Studio Romantic Song", 250, "Momina Mustehsan & Asim Azhar", 5)
dvd = DVD("Jawani Phir Nahi Ani", "Comedy Blockbuster Movie", 950, "PG-13", 165, ["Humayun Saeed", "Ahmed Ali Butt", "Vasay Chaudhry"])

```

```
book = Book("Jannat Kay Pattay", "Famous Urdu Novel", 800, "Nimra Ahmed", 750,
"Romantic Fiction")

# Display and call methods
mp3.viewFullDescription()
mp3.play()

dvd.viewFullDescription()
dvd.viewTrailer()

book.viewFullDescription()
book.previewContent()
```

OUTPUT:

```
PS C:\Users\HP\Documents\New folder>
Item Name: Tera Woh Pyar
Description: Coke Studio Romantic Song
Price: 250
Playing: Tera Woh Pyar by Momina Mustehsan & Asim Azhar
Item Name: Jawani Phir Nahi Ani
Description: Comedy Blockbuster Movie
Price: 950
Viewing trailer of: Jawani Phir Nahi Ani
Actors: Humayun Saeed, Ahmed Ali Butt, Vasay Chaudhry
Item Name: Jannat Kay Pattay
Description: Famous Urdu Novel
Price: 800
Previewing book: Jannat Kay Pattay by Nimra Ahmed
PS C:\Users\HP\Documents\New folder>
```

CONCLUSION:

In this lab, we practiced Python's object-oriented programming concepts such as **classes, objects, inheritance, and methods**. We developed programs like **Employee Management, Circle calculation, Power function, and Item inheritance (MP3, DVD, Book)**. This lab improved our understanding of **code reusability, structure, and OOP principles** in Python.
